Claude Code on a Pi

How to install Claude Code on a Raspberry Pi node, sign it in, and run it as a service so the Claude apps can reach the node through Remote Control. Two paths: the 44helper Claude Remote tab, or the same steps by hand.

NODE.JS 22 | CLAUDE CODE | SIGN IN | REMOTE CONTROL SERVICE

GUIDE · FROM THE 44HELPER STEPS

Source: the Claude Remote step tables in 44helper_v0_1_17_20261103.py on branch ccr-9d4fe19f-ssud83. Versions are the ones 44helper pins; a later 44helper may pin newer ones.

What you need

NEED	WHY
A 64-bit OS (aarch64) on the Pi, or an x86_64 machine	The Node.js download is for these two only. On a 32-bit OS (`armv7l`) the install stops with "No Node.js build for armv7l here". On a Pi Zero 2 W, install the 64-bit Raspberry Pi OS (Bookworm) or Debian 13 (Trixie).
Internet access from the Pi	Node.js comes from nodejs.org and Claude Code from the npm registry. The sign-in uses Claude's servers.
A Claude account that can use Claude Code	You sign in with the same account you use in the Claude apps. Remote Control then shows the Pi in those apps.
A normal login user on the Pi (not root)	Claude Code runs as this user. 44helper's form suggests `ubuntu`; on Raspberry Pi OS it is the user you created (often `pi`). Use your own.
A second device with a browser	To open the sign-in link.
About 200 MB of free disk, and some free memory	Node.js and Claude Code are installed under `/opt` and `/usr/local`. A Pi Zero 2 W has 512 MB of memory, so stop heavy jobs (or use Quiet System in instmon) while you install.

Check the processor and OS first:

```
uname -m                    # must say aarch64 (or x86_64)
grep -E '^(PRETTY_NAME|VERSION_CODENAME)=' /etc/os-release
```

With the 44helper Claude Remote tab

Open 44helper (port 9997), log in with the root password, and open the **Claude Remote** tab. Run the steps in order. Each Run button shows its command first; 44helper runs only the command in its table.

STEP	BUTTON	WHAT HAPPENS
1	Check what is here (read-only)	Shows the OS, the processor, any Node.js and Claude Code already installed, and the service state. Changes nothing.
2	Install Node.js 22	Downloads Node.js v22.23.4 for this processor, checks it against a SHA-256 pinned in 44helper, unpacks it to `/opt/node-v22.23.4`, and links `node`, `npm` and `npx` into `/usr/local/bin`. The Debian packages are too old for Claude Code, so they are not used.
3	Install Claude Code	Installs `@anthropic-ai/claude-code@2.1.317` with that Node.js and links `claude` into `/usr/local/bin`.
4	Sign in (manual)	Done in a terminal (Part 4). Use the **Terminal** button in the 44helper header, or SSH.
5	Create the Remote Control folder	Fill in the login user and the folder (default `~/claude-remote`), then run. The folder must be inside the user's home; the home itself and hidden folders are refused.
6	Start Claude Remote as a service	Writes `/etc/systemd/system/claude-remote.service`, enables it at boot, starts it, and shows its first log lines. It needs step 4 done first.
7	Check the service	Shows the status and the last 30 log lines.
8	Stop Claude Remote	Stops the service and disables it at boot. The Pi leaves the Claude apps.

Steps 5 and 6 use the same two fields. Give the user that signed in at step 4, and the same folder.

The same steps by hand

These are the commands from the 44helper table, laid out for a shell. Run them as your login user; `sudo` is used where root is needed. Set the two names first.

```
U=pi                      # your login user
D=/home/$U/claude-remote  # the Remote Control folder
```

3.1 Install Node.js 22 (pinned and checked)

```
a=$(uname -m)
if [ "$a" = aarch64 ]; then
  f=linux-arm64
  s=6287aa46b6c375d816e63ae66d024fbbfd70404825aabdfdf898a970871d2f4b
elif [ "$a" = x86_64 ]; then
  f=linux-x64
  s=1805bcd35e6815559bd2eb97db438d3fdb0eb3a5e6a977b7c1badca76337e9ce
else
  echo "No Node.js build for $a"
fi
cd /tmp
curl -fsSL -o node-cr.tar.xz \
  https://nodejs.org/dist/v22.23.4/node-v22.23.4-$f.tar.xz
echo "$s  node-cr.tar.xz" | sha256sum -c -   # must print: OK
sudo tar -xJf node-cr.tar.xz -C /opt
sudo mv /opt/node-v22.23.4-$f /opt/node-v22.23.4
rm -f node-cr.tar.xz
sudo ln -sf /opt/node-v22.23.4/bin/node /usr/local/bin/node
sudo ln -sf /opt/node-v22.23.4/bin/npm  /usr/local/bin/npm
sudo ln -sf /opt/node-v22.23.4/bin/npx  /usr/local/bin/npx
node --version                                  # v22.23.4
```

If `sha256sum` does not print OK, stop: the download is not the file 44helper was written for.

3.2 Install Claude Code

```
sudo /opt/node-v22.23.4/bin/npm install -g \
  @anthropic-ai/claude-code@2.1.317
sudo ln -sf /opt/node-v22.23.4/bin/claude /usr/local/bin/claude
claude --version
```

3.3 Create the folder, then sign in (Part 4)

```
mkdir -p "$D"
```

3.4 Run Remote Control as a service

```
sudo tee /etc/systemd/system/claude-remote.service >/dev/null <<EOF
[Unit]
Description=Claude Code remote control (44helper)
After=network-online.target
Wants=network-online.target

[Service]
Type=simple
User=$U
Environment=HOME=/home/$U
WorkingDirectory=$D
ExecStart=/usr/local/bin/claude remote-control
Restart=on-failure
RestartSec=10

[Install]
WantedBy=multi-user.target
EOF
sudo systemctl daemon-reload
sudo systemctl enable --now claude-remote
systemctl is-active claude-remote
journalctl -u claude-remote -n 15 --no-pager
```

This is the same unit 44helper writes, so 44helper's Check and Stop steps work with it.

Sign in once, as the login user

The service cannot sign in by itself. Sign in once in a terminal; the sign-in is kept in the user's home folder and the service uses it.

1. Open a terminal as the login user: the 44helper **Terminal** button (then `sudo -u pi -H bash` if it opened as root), or SSH in as that user.
2. Start Claude in the Remote Control folder:

   ```
   cd ~/claude-remote && claude
   ```

   44helper's step 4 does the same as root for you: `sudo -u ubuntu -H bash -c 'mkdir -p ~/claude-remote && cd ~/claude-remote && exec /usr/local/bin/claude'`, with your user in place of `ubuntu`.
3. Claude shows a sign-in link. Open it on your phone or computer, sign in with the same account as your Claude apps, and paste the code it gives you back into the terminal.
4. When Claude asks whether you trust the folder, answer **Yes**. The service will not start in a folder that is not trusted.
5. Leave Claude with `/exit` (or Ctrl+C twice).
6. Now run step 6 (Path A) or Part 3.4 (Path B).

Is it working?

```
systemctl status claude-remote --no-pager -l | head -25
journalctl -u claude-remote -n 30 --no-pager
```

- **Active (running)** and no errors in the log: the Pi should appear in the Claude apps under Remote Control.
- To work on the Node Controller code itself, clone the repository into the folder (`cd ~/claude-remote && git clone https://github.com/kd8pgk/ASL-DVS`). Claude Code reads the repository's `CLAUDE.md`, so the repo rules (versions, no comments, the testing policy) apply to its work.
- You can also use Claude Code directly over SSH: `cd ~/claude-remote && claude`.

5.1 If it does not start

SYMPTOM	CAUSE AND FIX
"No Node.js build for armv7l"	The OS is 32-bit. Install a 64-bit OS.
`sha256sum` **reports FAILED**	The download changed or was cut off. Delete it and try again; if it still fails, wait for a 44helper release with a new pin.
The service stops soon after start	Usually the sign-in was not done, was done as another user, or the folder was not trusted. Repeat Part 4 as the user named in the unit, in the same folder.
"Folder missing" from step 6	Run step 5 first, with the same user and folder.
The log says it cannot reach Claude	Check the Pi's internet access and clock (the 44helper Boot tab shows clock sync).
The install is slow or killed on a Pi Zero 2 W	Memory is short. Stop other jobs (instmon Quiet System) and run the step again.

What this gives access to

- **Who can reach it.** Remote Control makes the Pi reachable from the Claude apps signed in to your account. Anyone with that account can direct Claude on the Pi. Protect the account.
- **What it can do.** Claude Code works with the rights of the login user, in the folder you chose. It is not run as root. If that user can use `sudo`, Claude can ask to run root commands; read each request before you allow it.
- **Secrets.** Do not keep passwords or keys in the Remote Control folder. Node Controller keys live in `/etc/asl_dvs`, outside it.
- **Turning it off.** Step 8 (or `sudo systemctl disable --now claude-remote`) removes the Pi from the apps at once.

Taking it off the Pi

The 44helper purge for this tab runs these, in order, then checks that nothing is left:

```
sudo systemctl disable --now claude-remote
sudo rm -f /etc/systemd/system/claude-remote.service
sudo systemctl daemon-reload
sudo /opt/node-v22.23.4/bin/npm uninstall -g @anthropic-ai/claude-code
sudo rm -f /usr/local/bin/claude
sudo rm -f /usr/local/bin/node /usr/local/bin/npm /usr/local/bin/npx
sudo rm -rf /opt/node-v22.23.4
```

The sign-in stays in the user's home folder (`~/.claude`). Delete that folder by hand if you want it gone too. The Remote Control folder and anything in it are kept.